



St Aloysius' College
Year 11 Preliminary Examination
2017

MATHEMATICS

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is the domain of the function $y = \sqrt{x+4}$?

(A) $y \geq -0$

(B) $y > 0$

(C) $x \geq -4$

(D) $x > -4$

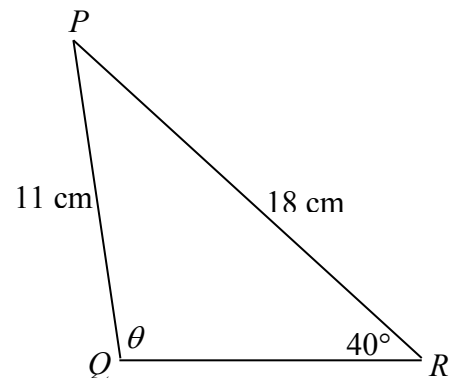
2 The diagram shows the triangle PQR where $PQ = 11$ cm, $PR = 18$ cm and $\angle PRQ = 40^\circ$. Which expression correctly gives the value of $\sin \theta$?

(A) $\frac{18}{11 \sin 40^\circ}$

(B) $\frac{11}{18 \sin 40^\circ}$

(C) $\frac{11 \sin 40^\circ}{18}$

(D) $\frac{18 \sin 40^\circ}{11}$



- 3 $A(3, y)$ and $B(7, 2)$ are points on the number plane. The gradient of AB is $m = -3$.

What is the value of y ?

- (A) 14
- (B) $\frac{1}{14}$
- (C) -10
- (D) $-\frac{1}{10}$

- 4 What are the solutions of the equation $|3x - 5| = 10$?

- (A) $x = -5, 15$
- (B) $x = -5, 5$
- (C) $x = -1\frac{2}{3}, 5$
- (D) $x = -5, 1\frac{2}{3}$

- 5 How many solutions are there to the equation $x^3 - x^2 - 5x = 0$?

- (A) 1
- (B) 3
- (C) 0
- (D) 2

6 Solve $\sin \theta = \frac{1}{\sqrt{2}}, 0 \leq \theta \leq 360^\circ$.

- (A) $\theta = 45^\circ, 135^\circ$
- (B) $\theta = 135^\circ, 225^\circ$
- (C) $\theta = 225^\circ, 315^\circ$
- (D) $\theta = 315^\circ, 45^\circ$

7 Simplify $\lim_{h \rightarrow 0} \frac{4(x+h)^2 - 4x^2}{h}$

- (A) $8h$
- (B) $8xh + 8h$
- (C) $8x + 8h$
- (D) $8x$

8 The line $y = mx - 2$ is a tangent to the curve $y = 3x^2 - 2x + 1$ at the point $(1, 2)$.

What is the value of m ?

- (A) 4
- (B) $\frac{1}{2}$
- (C) -4
- (D) 2

- 9 The function $y = f(x)$ is defined as:

$$f(x) = \begin{cases} x^2 + 1 & x < 0 \\ 1 - x^2 & x \geq 0 \end{cases}$$

What is the range of $f(x)$?

- (A) $y \leq -1, y \geq 1$
 - (B) all real y
 - (C) $y < -1, y \geq 1$
 - (D) $-1 \leq y \leq 1$
- 10 If $8^{x+3} \times 2^{x-2} = 2^x \times 4^{3x-1}$, what is the value of x ?
- (A) 0
 - (B) 2
 - (C) 3
 - (D) -3

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Question 11 (15 marks) **Use a SEPARATE writing booklet.**

- (a) Evaluate $\sqrt{\frac{2.6^2 + 5.2^2}{1.3}}$ correct to 2 significant figures. 2
- (b) Expand and simplify $5(x+3)^2 - x(7-2x)$. 2
- (c) Factorise $3x^2 + 15x - 72$. 2
- (d) Rationalise the denominator of the expression $\frac{4 + \sqrt{6}}{\sqrt{2}}$. 2
- (e) The point $M(-4, 1)$ is the midpoint of the points $A(2, 6)$ and $B(k, -4)$. 1
Find the value of k .
- (f) Determine if the function $f(x) = 3x^4 - 4x^2$ is even, odd or neither. 2
- (g) Find the equation of the circle with radius 3 units and centre $(0, -4)$. 1
- (h) Shade the region defined by $y \geq x^2 + 1$, $x \geq 0$ and $y \leq 3$. Show all y -intercepts and points of intersection. 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Differentiate the following functions with respect to x

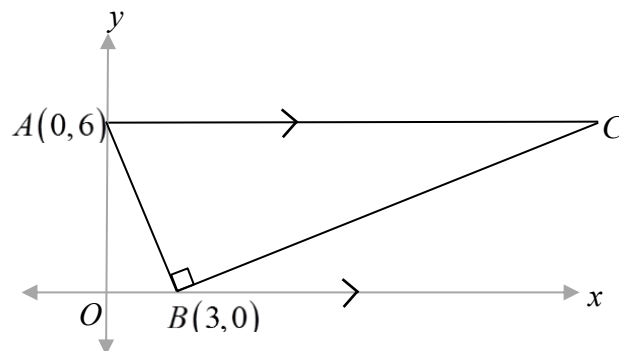
(i) $f(x) = 4x^2 - 6x - 9$ 1

(ii) $f(x) = \frac{x}{\sqrt{x}}$ 2

(iii) $f(x) = \frac{2x^3}{3+x^2}$ 2

(b) Differentiate from first principles $f(x) = 2x^2 - 3x + 7$ 2

(c) The diagram shows the triangle ABC . The coordinates of A are $(0,6)$ and the coordinates of B are $(3,0)$. BC is perpendicular to AB . AC is parallel to the x -axis.



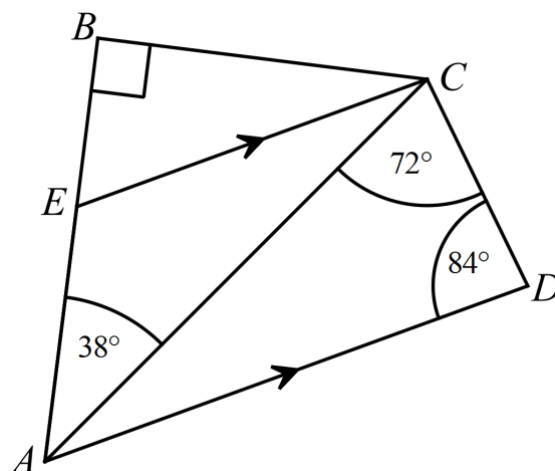
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SCALE**

- (i) Find the gradient of line BC . 1
- (ii) Show that the equation of the line BC is $x - 2y - 3 = 0$ 1
- (iii) Hence or otherwise, show that the coordinates of the point C are $(15, 6)$. 1
- (iv) Find the length of BC . Express your answer in exact simplified form. 2
- (v) Find the size of $\angle ACB$, correct to the nearest degree. 2
- (vi) Find the area of triangle ABC . 1

End of Question 12

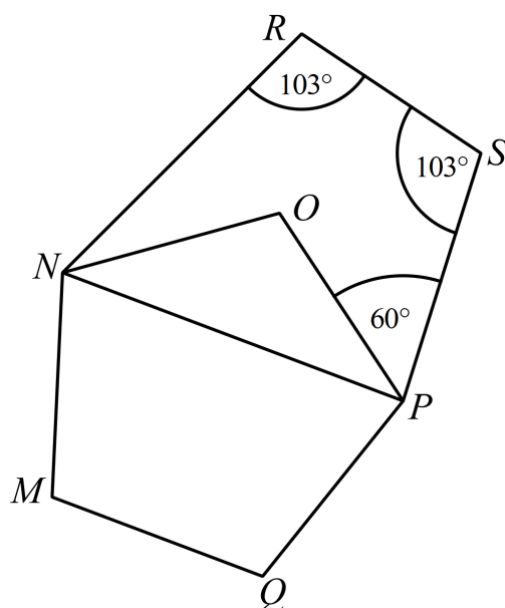
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) If $f(x) = \begin{cases} 1 & x < 0 \\ x+2 & x \geq 0 \end{cases}$
- (i) Evaluate $f(-4) + f\left(-\frac{1}{2}\right) + f(0)$ 2
- (ii) Sketch $y = f(x)$ for $-3 \leq x \leq 3$ 2
- (b) Given $\sin \theta = -\frac{4}{5}$ and $\tan \theta > 0$, find the exact value of
- (i) $\cos \theta$ 1
- (ii) $\cot \theta$ 1
- (c) Prove $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$ 3
- (d) Find the gradient of the normal to the curve $f(x) = (2x + 3)^2$ at the point where $x = -1$. 2
- (e) $ABCD$ is a quadrilateral, with the diagonal AC drawn as shown. E is a point on AB , such that $EC \parallel AD$. Find the size of $\angle AEC$ and $\angle BCE$. 2



Question 13 Continues on Page 9

- (f) The polygon $MNOPQ$ is a regular pentagon $\angle OPS = 60^\circ$, $\angle RSP = 103^\circ$ and $\angle NRS = 103^\circ$ as shown in the diagram.



- | | | |
|------|---------------------------------|----------|
| (i) | Find the size of $\angle ONP$. | 1 |
| (ii) | Find the size of $\angle ONR$. | 1 |

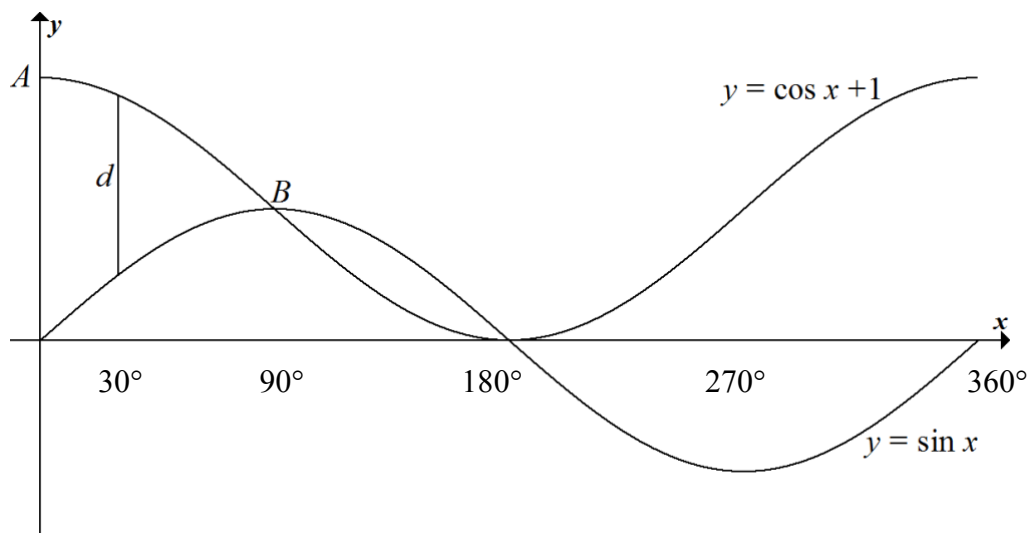
End of Question 13

Question 14 (15 marks) Use a **SEPARATE** writing booklet.

(a) Solve $\cos^2 \theta = \frac{1}{4}$ where $0 \leq \theta \leq 360^\circ$. 2

(b) Solve for x , $\log_5 \left(\frac{x+10}{x-2} \right) = 2$. 2

(c) The diagram shows the graphs $y = \sin x$ and $y = \cos x + 1$ for $0^\circ \leq x \leq 360^\circ$.



- (i) State the coordinates of the point A . 1
- (ii) Determine the y -ordinate of the point B . 1
- (iii) Find the coordinates of the first point of intersection of the two graphs for $x \geq 360^\circ$. 1
- (iv) Let d be the vertical distance between the two graphs when $x = 30^\circ$. Find the exact value of d . 1
- (v) Find the next value of x where the vertical distance between the two graphs equals d . 1

Question 14 Continues on Page 11

(d) A yacht sails 640 metres from point P to point A on a bearing of 050° . It then sails 960 metres from point A to point B on a bearing of 120° .

- (i) Draw a clear diagram showing the above information. **2**
- (ii) Find the distance of point B from point P correct to the nearest metre. **2**
- (iii) Find the bearing of point P from point B correct to the nearest degree. **2**

End of Examination

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$$\sin \theta = \frac{18 \sin 40^\circ}{11}$$

④

$$-3 = \frac{y-2}{3-7}$$

$$-3 = \frac{y-2}{-4}$$

$$12 = y - 2$$

$$\therefore y = 14$$

A

$$3x - 5 = 10$$

$$3x = 15$$

$x = 5$

$$3x - 5 = -10$$

$$3x = -5$$

$$x = -\frac{5}{3}$$

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$x = 0$

$$x = -b \pm \sqrt{b^2 - 4ac}$$

$$= 1 \pm \sqrt{1+20}$$

$$= \frac{1 \pm \sqrt{21}}{2}$$

B

3 solutions

Q6 $\sin \theta = \frac{1}{\sqrt{2}}$



$\therefore \theta = 45^\circ, 135^\circ$

(A)

Q7 $\lim_{h \rightarrow 0} \frac{4(x^2 + 2xh + h^2) - 4x^2}{h}$

$= \lim_{h \rightarrow 0} \frac{\cancel{4x^2} + 8xh + 4h^2 - \cancel{4x^2}}{h}$

$= \lim_{h \rightarrow 0} \frac{\cancel{h}(8x + 4h)}{\cancel{h}}$

$= \lim_{h \rightarrow 0} 8x + 4h$

$= 8x$

(D)

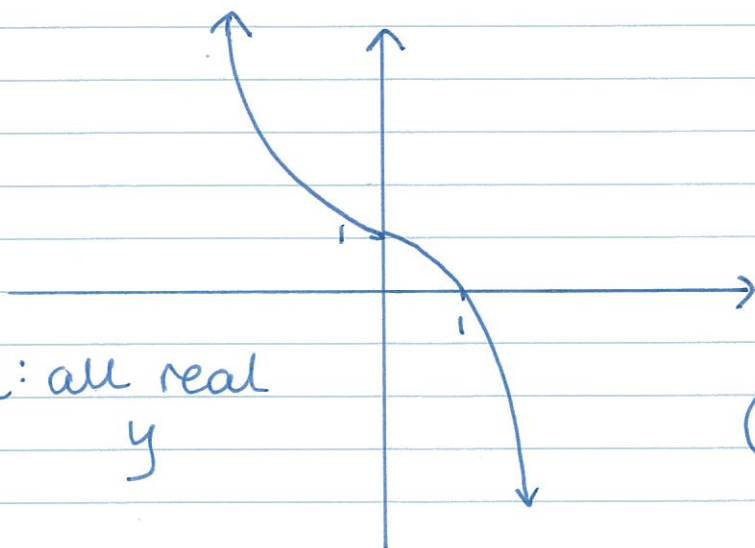
Q8 $y' = 6x - 2 \quad (1, 2)$

$m = 6(1) - 2$

$m = 4$

(A)

Q9



$\mathbb{R}$: all real
y

(B)

$$Q10 \quad 8^{x+3} \times 2^{x-2} = 2^x \times 4^{3x-1}$$

$$2^{3(x+3)} \times 2^{x-2} = 2^x \times 2^{2(3x-1)}$$

$$2^{3x+9+x-2} = 2^{x+6x-2}$$

$$2^{4x+7} = 2^{7x-2}$$

$$4x+7 = 7x-2$$

$$9 = 3x$$

$$\therefore x = 3$$

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SECTION II

Question 11

$$a) = 5.0990... \\ = 5.1$$

$$b) = 5(x^2 + 6x + 9) - 7x + 2x^2 \\ = 5x^2 + 30x + 45 - 7x + 2x^2 \\ = 7x^2 + 23x + 45$$

$$c) = 3(x^2 + 5x - 24) \\ = 3(x+8)(x-3)$$

$$d) = \frac{4+\sqrt{6}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ = \frac{4\sqrt{2} + \sqrt{12}}{2} \\ = \frac{4\sqrt{2} + 2\sqrt{3}}{2} = 2\sqrt{2} + \sqrt{3}$$

$$e) M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\therefore -4 = \frac{2+k}{2}$$

$$-8 = 2+k$$

$$k = -10$$

$$f) f(x) = 3x^4 - 4x^2$$

$$\begin{aligned} f(-x) &= 3(-x)^4 - 4(-x)^2 \\ &= 3x^4 - 4x^2 \\ &= f(x) \end{aligned}$$

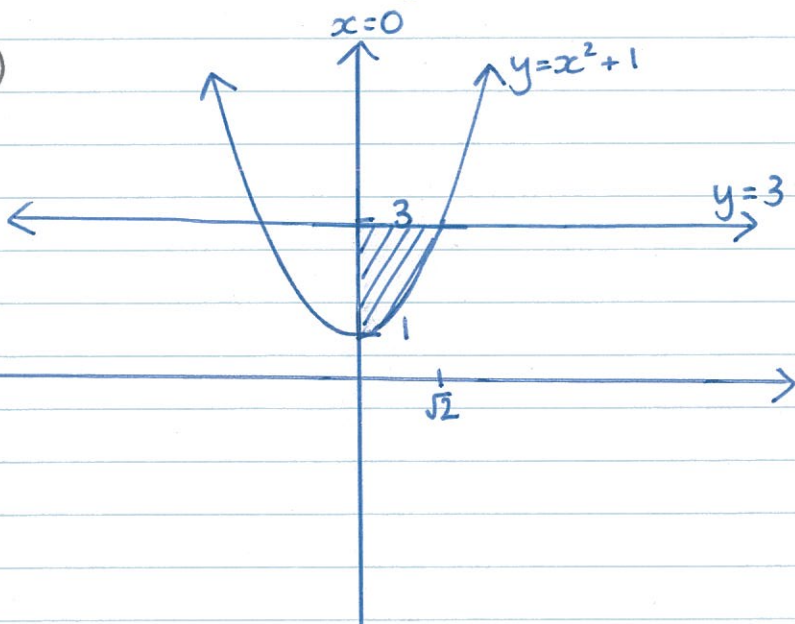
$\therefore$ even

$$g) (x-h)^2 + (y-k)^2 = r^2$$

$$\therefore (x-0)^2 + (y+4)^2 = 9$$

$$x^2 + (y+4)^2 = 9$$

h)



Question 12

a) i. $f'(x) = 8x - 6$

ii. $f(x) = x \cdot x^{-\frac{1}{2}}$
 $= x^{\frac{1}{2}}$

$$f'(x) = \frac{1}{2} \cdot x^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x}}$$

iii. $f'(x) = \frac{6x^2(3+x^2) - 2x^3(2x)}{(3+x^2)^2}$

$$= \frac{18x^2 + 6x^4 - 4x^4}{(3+x^2)^2}$$

$$= \frac{18x^2 + 2x^4}{(3+x^2)^2}$$

b) $f(x) = 2x^2 - 3x + 7$

$$\begin{aligned} f(x+h) &= 2(x+h)^2 - 3(x+h) + 7 \\ &= 2(x^2 + 2xh + h^2) - 3x - 3h + 7 \\ &= 2x^2 + 4xh + 2h^2 - 3x - 3h + 7 \end{aligned}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 3x - 3h + 7 - (2x^2 - 3x + 7)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 3h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(4x + 2h - 3)}{h}$$

$$= \lim_{h \rightarrow 0} 4x + 2h - 3$$

$$= 4x - 3$$

$$\begin{aligned}
 \text{c) i. } m_{AB} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{0 - 6}{3 - 0} \\
 &= -2
 \end{aligned}$$

$$AB \perp AC \therefore m_{AB} \times m_{AC} = -1$$

$$\therefore m_{AC} = \frac{1}{2}$$

$$\text{ii. } y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{1}{2}(x - 3)$$

$$\begin{aligned}
 2y &= x - 3 \\
 \therefore x - 2y - 3 &= 0
 \end{aligned}$$

$$\text{iii. } y = 6, \quad x - 2y - 3 = 0$$

$$\begin{aligned}
 x - 12 - 3 &= 0 \\
 x &= 15
 \end{aligned}$$

$$\therefore C(15, 6)$$

$$\begin{aligned}
 \text{iv. } d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(15 - 3)^2 + (6 - 0)^2} \\
 &= \sqrt{144 + 36} \\
 &= \sqrt{180} \\
 &= 6\sqrt{5}
 \end{aligned}$$

$$\begin{aligned}
 \text{v. } d_{AB} &= \sqrt{(3 - 0)^2 + (0 - 6)^2} \\
 &= \sqrt{45} \\
 &= 3\sqrt{5}
 \end{aligned}$$

$$\tan \theta = \frac{AB}{BC}$$

$$\tan \theta = \frac{3\sqrt{5}}{6\sqrt{5}}$$

$$\tan \theta = \frac{1}{2}$$

$$\theta = 27^\circ$$

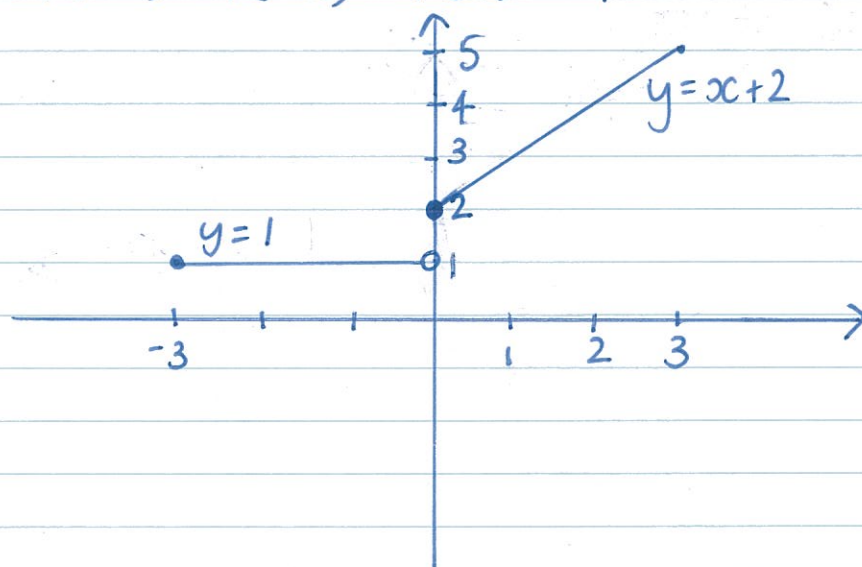
$$\begin{aligned}
 \text{vi. } A &= \frac{1}{2}bh \\
 &= \frac{1}{2} \times 3\sqrt{5} \times 6\sqrt{5} \\
 &= 45u^2
 \end{aligned}$$

Question 13

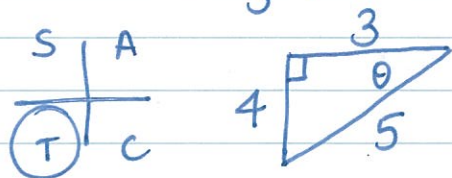
$$\begin{aligned}
 \text{a) i. } f(-4) &= 1 \\
 f\left(-\frac{1}{2}\right) &= 1 \\
 f(0) &= 0+2 \\
 &= 2
 \end{aligned}$$

$$\therefore f(-4) + f\left(-\frac{1}{2}\right) + f(0) = 4$$

ii.



$$\text{b) } \sin \theta = -\frac{4}{5}, \tan \theta > 0$$



$$\text{i. } \cos \theta = -\frac{3}{5}$$

$$\text{ii } \cot \theta = \frac{3}{4}$$

$$c) \frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A} = 2 \operatorname{cosec} A$$

$$\text{LHS} = \frac{\sin^2 A}{(1+\cos A)\sin A} + \frac{(1+\cos A)^2}{(1+\cos A)\sin A}$$

$$= \frac{\sin^2 A + 1 + 2\cos A + \cos^2 A}{(1+\cos A)\sin A}$$

$$= \frac{1 + 1 + 2\cos A}{(1+\cos A)\sin A}$$

$$= \frac{2(1+\cos A)}{(1+\cos A)\sin A}$$

$$= 2 \operatorname{cosec} A$$

$$= \text{RHS}$$

$$d) f'(x) = 2(2x+3) \times 2$$

$$f'(-1) = 2(2(-1)+3) \times 2$$
$$= 4$$

$$\therefore m_T = 4$$

$$m_T \times m_N = -1$$

$$\therefore m_N = \frac{-1}{4}$$

e) $\angle CAD + 72^\circ + 84^\circ = 180^\circ$ (angle sum of a triangle is 180°)

$$\angle CAD = 24^\circ$$

$$\angle CAD = \angle ACE \text{ (alternate angles on parallel lines are equal)}$$

$\angle AEC + 38^\circ + 24^\circ = 180^\circ$ (angle sum of a triangle is 180°)

$$\underline{\angle AEC = 118^\circ}$$

$$\angle BEC + 118^\circ = 180^\circ \text{ (angles on a straight line are supplementary)}$$

$$\angle BEC = 62^\circ$$

$$62^\circ + 90^\circ + \angle BCE = 180^\circ \text{ (angle sum of a triangle is } 180^\circ)$$

$$\underline{\angle BCE = 28^\circ}$$

f) i. $\angle PON = \frac{180^\circ(n-2)}{n}$ (interior angle of a regular polygon)
 $= 108^\circ$

$$\angle ONP = \frac{180^\circ - 108^\circ}{2} \text{ (base angles of an isosceles triangle are equal)}$$

$$\angle ONP = 36^\circ$$

ii. reflex $\angle PON = 360^\circ - 108^\circ$ (angles at a point add to 360°)
 $= 252^\circ$

$$252^\circ + 60^\circ + 103^\circ + 103^\circ + \angle ONR = 540^\circ \text{ (angle sum of a pentagon is } 540^\circ)$$

$$\therefore \angle ONR = 22^\circ$$

c) i. A (0, 2)

ii. $x = 90^\circ$

$$y = \sin(90^\circ)$$

$$y = 1$$

iii. (450°, 1)

iv. $y = \cos 30^\circ + 1$

$$y = \sin 30^\circ$$

$$\therefore d = \cos 30^\circ + 1 - \sin 30^\circ$$

$$= \frac{\sqrt{3}}{2} + 1 - \frac{1}{2}$$

$$= \frac{\sqrt{3} + 2 - 1}{2}$$

$$= \frac{\sqrt{3} + 1}{2}$$

v. $y = \cos 240^\circ + 1 - \sin 240^\circ$

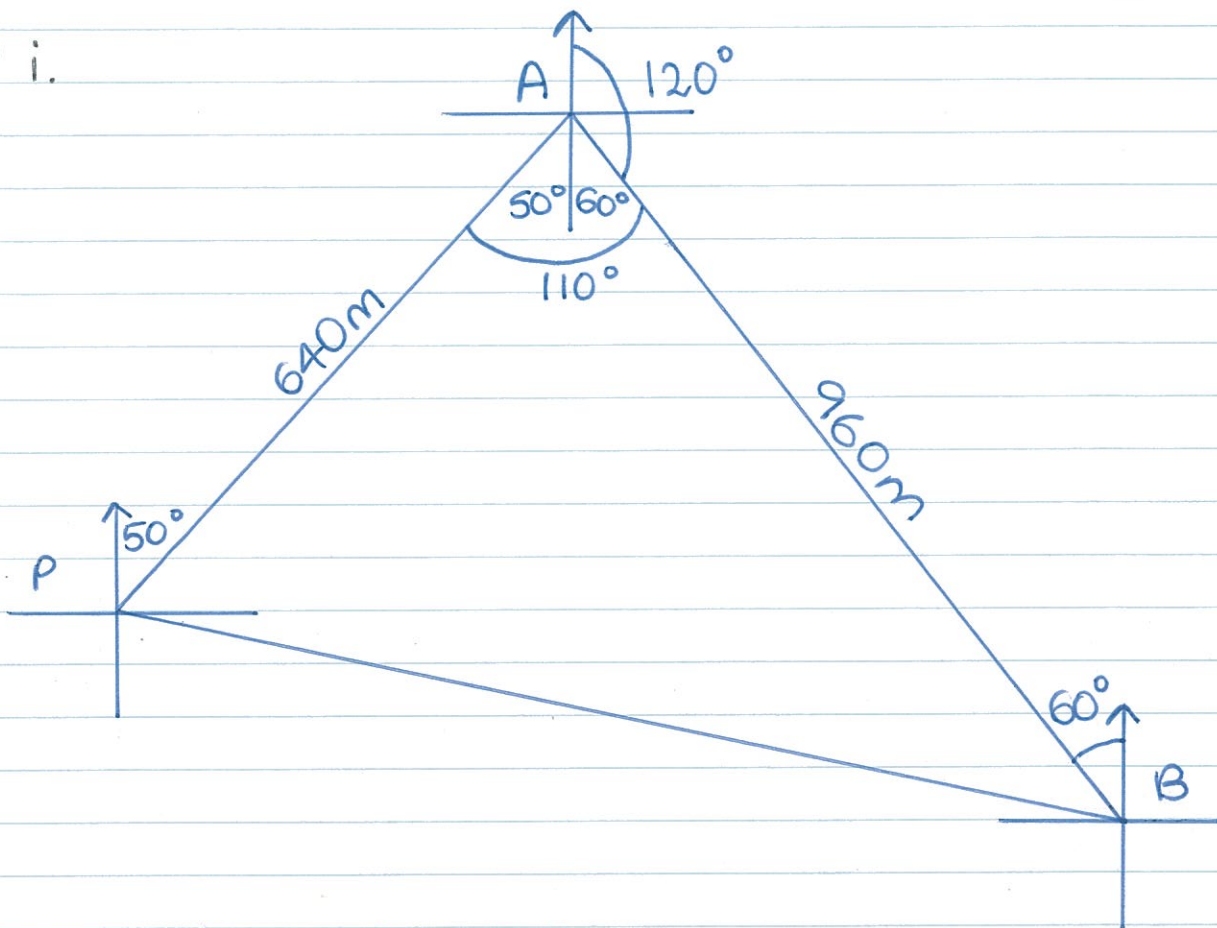
$$= -\frac{1}{2} + 1 + \frac{\sqrt{3}}{2}$$

$$= \frac{-1 + 2 + \sqrt{3}}{2}$$

$$= \frac{1 + \sqrt{3}}{2}$$

$\therefore$ next value of x is 240° .

d) i.



$$\text{ii. } (PB)^2 = 640^2 + 960^2 - 2 \times 640 \times 960 \cos 110^\circ$$

$$PB = 1323.4...$$

$$PB = 1323 \text{ m}$$

$$\text{iii. } \frac{\sin \angle APB}{640} = \frac{\sin 110^\circ}{1323}$$

$$\sin \angle ABP = \frac{640 \sin 110^\circ}{1323}$$

$$\begin{aligned} \angle ABP &= 27.0...^\circ \\ &= 27^\circ \end{aligned}$$

$$\therefore \text{Bearing of P from B} = 360^\circ - 27^\circ - 60^\circ = 273^\circ$$